Homework 1

Write a decoder for the following instruction set. The design will be fully combinational.

31 30 29 28 27 26 25	24 23 22 21 20	19 18 17 16 15 14 13	12 11 10 9 8 7 6 5	4 3 2 1 0	
imm[31:20]		1110101	imm[19:12]	rd	LUI
imm[31:20]		1110100	imm[19:12]	rd	AUIPC
imm[20|10:1|11]		1101111	imm[19:12]	rd	JAL
imm[11:0]		1100111	rs1 000	rd	JALR
imm[12|10:5]	rs2	1100011	rs1 100	imm[4:1|11]	BEQ
imm[12|10:5]	rs2	1100011	rs1 101	imm[4:1|11]	BNE
imm[12|10:5]	rs2	1100011	rs1 000	imm[4:1|11]	BGEU
imm[12|10:5]	rs2	1100011	rs1 001	imm[4:1|11]	BLTU
imm[12|10:5]	rs2	1100011	rs1 010	imm[4:1|11]	BGE
imm[12|10:5]	rs2	1100011	rs1 011	imm[4:1|11]	BLT
imm[11:0]		1100000	rs1 000	rd	LBU
imm[11:0]		1100000	rs1 001	rd	LHU
imm[11:0]		1100000	rs1 100	rd	LB
imm[11:0]		1100000	rs1 101	rd	LH
imm[11:0]		1100000	rs1 110	rd	LW
imm[11:5]	rs2	1100001	rs1 000	imm[4:0]	SB
imm[11:5]	rs2	1100001	rs1 001	imm[4:0]	SH
imm[11:5]	rs2	1100001	rs1 010	imm[4:0]	SW
imm[11:0]		1100100	rs1 000	rd	ADDI
imm[11:0]		1100100	rs1 010	rd	SLTI
imm[11:0]		1100100	rs1 011	rd	SLTIU
imm[11:0]		1100100	rs1 100	rd	ORI
imm[11:0]		1100100	rs1 110	rd	XORI
imm[11:0]		1100100	rs1 111	rd	ANDI
0000000	imm[4:0]	1100100	rs1 001	rd	SLLI
0000000	imm[4:0]	1100100	rs1 101	rd	SRLI
0000010	imm[4:0]	1100100	rs1 101	rd	SRAI
0000000	rs2	1110001	rs1 000	rd	ADD
0000010	rs2	1110001	rs1 000	rd	SUB
0000000	rs2	1110001	rs1 001	rd	SLL
0000000	rs2	1110001	rs1 010	rd	SLT
0000000	rs2	1110001	rs1 011	rd	SLTU
0000000	rs2	1110001	rs1 100	rd	XOR
0000000	rs2	1110001	rs1 101	rd	SRL
0000010	rs2	1110001	rs1 101	rd	SRA
0000000	rs2	1110001	rs1 110	rd	OR
0000000	rs2	1110001	rs1 111	rd	AND

The mentioned instruction set will be designed inside the src/decoder.sv file in the given repository, using the struct definitions provided in src/pkg/typedef_pkg.sv.

Information about the purpose of the signals in the struct is given below:

valid: Indicates that the information is ready to be used. It is a very common signal. It exists in almost every design to enable/disable the system or to stall operations.
is_mem: Indicates that the operation is a memory operation. (LBU, LHU, LB, LH, LW, SB, SH, SW)
is_pc_changer: Indicates that the operation will cause an abnormal change in the program counter. (JAL, JALR, BEQ, BNE, BGEU, BLTU, BGE, BLT)
rd_valid: Indicates that the rd field in the instruction (instr[4:0]) should be processed as a valid destination register.
rs1_valid: Similar to rd_valid, but for the rs1 field.
rs2_valid: Similar to rd_valid, but for the rs2 field.
rd_idx: The rd field is extracted and provided separately for easier processing. Its validity is determined by rd_valid.
rs1_idx: Similar to rd_idx, but for rs1.
rs2_idx: Similar to rd_idx, but for rs2.
imm_valid: Indicates that the immediate signal is valid.
imm: The properly arranged immediate value formed from the related parts of the instruction. In RV32I, all immediates are sign-extended except for the shift-immediate instructions (SLLI, SRLI, SRAI), where the immediate represents a zero-extended shift amount.
op: As can be seen in the op_e struct, the instruction is decoded and assigned a name. This makes it easier to trace instructions and simplifies coding in the related execution units.

Files allowed to modify (DO NOT change any other file except them)
- src/decoder.sv
- tb/tb_decoder.sv (it's just for you to test your code)

How to test code:
In tb_decoder.sv, you will see the instruction mapping at lines between 397 and 469. You must fill in that mapping to test your code. You can find detailed explanations in the code.

Kindly Reminder:
Since this is the first assignment, we wanted to start with something simple. Additionally, since some of you are new to hardware description languages, we thought it would be beneficial to start with an assignment that incorporates combinational logic.

As a beneficial reminder, we recommend avoiding the use of artificial intelligence tools. While you have time, it is important to gain as much competence as possible in different topics. This is one of the main reasons why this course is offered. After this assignment, the upcoming ones will be more difficult.